Junxi Yi

Ph.D. Student, Bioengineering (UIUC)

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Education

2022–Present — **Ph.D. in Bioengineering**
University of Illinois at Urbana-Champaign (UIUC)
Advisor: Yun-Sheng Chen, Ph.D.

2018–2022 — **B.Sc. in Chemical Biology**
University of Science and Technology of China (USTC)
Lu Jiaxi Chemistry Honors Program.

Honors & Awards

- Beckman Institute Graduate Fellow, UIUC (2026)
- Mavis Future Faculty Fellow, UIUC (2026)
- Award for Outstanding Overseas Study Elite, China Scholarship Council (2026)
- Lu Jiaxi Talent Program Scholarship, USTC (2018, 2020, 2021)
- Wu Xuezhou Scholarship, USTC & Changchun Institute of Applied Chemistry, CAS (2020)
- Outstanding Student Scholarship (Bronze), USTC (2021)

Publications

* *indicates co-first authors.*

Manuscripts under review and in preparation

1. H. Li, S. Zhao, F. Xie, **J. Yi**, B. Chen, M. Chauhan, M. Oelze, K. Li, J. S. Moore, Y.-S. Chen. "An Open-Source Automated Focused Ultrasound System for Programmable Mechanophore Screening." *In preparation.*

Peer-reviewed journals

9. M. Chauhan, S. Kumar, **J. Yi**, Y.-S. Chen. "Engineering Nano-Immune Interfaces for Precision STING Activation: From Molecular Engineering to Cancer Immunotherapy." **ACS Nano Medicine** (2026). DOI: 10.1021/acsnanomed.6c00203.

8. **J. Yi***, F. Xie*, J. Q. Moller*, Z. Xu, S. Zhao, Y. Diao, A. D. B. Leakey, J. S. Moore, Y.-S. Chen. "Ultrasound-driven mechanophore activation in living plants." **Proceedings of the National Academy of Sciences of the United States of America** (2026). DOI: 10.1073/pnas.2533066123.

7. S. Zhao, **J. Yi**, Y. Qiu, Z. Huang, Y.-S. Chen. "Photo-activated ultrasound localization imaging with laser-activated nanodroplets." **Communications Engineering** (2026). DOI: 10.1038/s44172-026-00592-w.

6. **J. Yi***, M. Chauhan*, Y. Zhao, Y.-S. Chen. "Near-infrared-triggered nanodroplets for macrophage-targeted cancer immunotherapy." **Small Methods** (2025). DOI: 10.1002/smtd.202401679. **Frontispiece paper**

5. **J. Yi**, Y.-S. Chen. "How can nanotechnology be leveraged to produce multifunctional contrast agents to improve imaging and diagnostics?" **Nanomedicine** (2025). DOI: 10.1080/17435889.2025.2501515.

4. S. Zhao S. Paul, **J. Yi**, Y.-S. Chen. "Dual modal photoacoustic/fast ultrasound localization imaging." **Bio-protocol** (2025). DOI: 10.21769/BioProtoc.5247.

3. J. Wang*, S. Zhao*, **J. Yi***, Y. Sun, M. Agrawal, M. L. Oelze, K. Li, J. Moore, Y.-S. Chen. "Injectable mechanophore nanoparticles for deep-tissue mechanochemical dynamic therapy." **ACS Nano** (2024). DOI: 10.1021/acsnano.4c04090.

2. X. Hu*, Y. Xu*, **J. Yi***, C. Wang, Z. Zhu, T. Yue, H. Zhang, X. Wang, F. Wu, L. Bai, H. Liu, Q. Chen. "Using protein design to monomerize a bright near-infrared fluorescent protein." **ACS Synthetic Biology** (2024). DOI: 10.1021/acssynbio.3c00643.

Conference publications

1. **J. Yi**, M. C. Fanning, M. R. Surburg, F. Buranits, P. W. Hsu, Y. Li, A. J. Loarte, A. Michailidis, Z. S. Rehman, M. Griebel, C. Cvetkovic. "WIP: An Undergraduate Neural Tissue Engineering Lab Course." **2026 ASEE Annual Conference & Exposition**, Charlotte, NC (2026). https://peer.asee.org/60818.

Conference Presentations

- **2026** — Poster: "Ultrasound-driven molecular activation in vivo." NSF cMUSIC Meeting, Raleigh, NC.
- **2025** — Poster: "Near-infrared-triggered nanodroplets for macrophage-targeted cancer immunotherapy." Cancer Center at Illinois Student Poster Session, Urbana, IL.
- **2024** — Poster: "Cancer killing effect of ultrasound-responsive mechanophore nanoparticles." Cancer Research Annual Meeting, San Diego, CA.
- **2023** — Poster: "In vivo mechanochemical dynamic cancer therapy using focused ultrasound-triggered mechanophores." BMES Annual Meeting, Seattle, WA.
- **2023** — Poster: "Non-invasive imaging of gut microbes with reversibly switchable photoacoustic protein." BMES Annual Meeting, Seattle, WA.

Invited Talks

- **2025** — Talk: "Near-infrared-triggered nanodroplets for macrophage-targeted cancer immunotherapy." Mechanochemical Dynamic Therapy Workshop, Cancer Center at Illinois, Urbana, IL.
- **2024** — Talk: "Injectable mechanophore nanoparticles for deep-tissue mechanochemical dynamic therapy." Bioengineering Graduate Research Symposium, University of Illinois at Urbana-Champaign, Urbana, IL.
- **2022** — Panel: "Flying Across the Ocean: panel discussion of overseas research experiences." Department of Environment, USTC, Hefei, Anhui, China.

Research Experience

2022–Present
University of Illinois at Urbana-Champaign

Research Assistant

- **Developed** ultrasound-responsive mechanophore nanocarriers and imaging-guided actuation strategies for controlled drug release and cancer therapy.
- **Built** bubble-nucleation and gas-vesicle integration methods to improve activation efficiency and localization; mentored an undergraduate researcher.

Jul–Oct 2021
The University of Edinburgh

Research Intern

- **Designed** plant E3 ubiquitin ligases to degrade pathogen effectors and improve plant immunity.

Jul–Oct 2020
McMaster University

Research Intern

- **Developed** a machine-learning method for molecular potential energy surface calculations.

Teaching Experience

Spring 2026
Department of Bioengineering, UIUC

Teaching Assistant, BIOE 486 (Applied Deep Learning in Biomedical Imaging)
Developed and graded assignments; supported PyTorch-based imaging pipelines in office hours.

Spring 2026; Fall 2025
Department of Bioengineering, UIUC

Teaching Assistant, BIOE 303 (Quantitative Physiology Lab)
Led lab/discussion sections; graded reports and provided feedback on analysis and writing.

Fall 2025
Department of Bioengineering, UIUC

Teaching Assistant, NE 431 (Neural Engineering Lab)
Supported lab instruction and troubleshooting; graded assignments and reports.

Spring 2025; Fall 2024
Department of Bioengineering, UIUC

Teaching Assistant, BIOE 210 (Linear Algebra for Biomedical Data Science)
Delivered lectures; built slides and MATLAB demos; hosted weekly office hours.

Fall 2023
Department of Bioengineering, UIUC

Teaching Assistant, BIOE 306 (Biofabrication)
Led 3D bioprinting/biomaterial fabrication labs; assisted with setup and data analysis.

Outreach & Service

June 2024
Urbana, IL

ECE Summer School
Led lab tours and delivered hands-on demonstrations of an ultrasound imaging system.

April 2023; April 2024 Urbana, IL	**Beckman Open House** Designed and presented public-facing demonstrations on 3D printing and ultrasound-based "treasure hunting".
July 2023 Urbana, IL	**Discover ECE MS Day Camp** Facilitated hands-on STEM activities for visiting students.
July 2020 Chongzuo, Guangxi, China	**Volunteer Instructor, "Zhuangmiao Program"** Delivered a two-week science/engineering curriculum for a rural primary-school cohort.